



Credit Card Fraud Detection: A Systematic Review

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Abstract. Due to the tremendous growth of technology, digitalization has become the key aspect in the banking sector. As online transaction increases, the fraud rate grows simultaneously. Even though many techniques are available to identify the fraudulent transaction, the fraudsters adapt their own paradigm. This review intends to present the research studies accomplished on Credit Card Fraud Detection (CCFD) by highlighting the challenge of class imbalance and the various Machine Learning techniques, it also extends the efficient evaluation metrics particularly for CCFD. As the dataset is more sensitive and less available we have outlined the web sources of available datasets and trending software tools used in the deployment of CCFD.

Keywords: Fraud detection · Class imbalance · Machine learning · Evaluation metrics

1 Introduction

Cashless transactions are the most popular method used to purchase products through e-commerce. Credit Cards are commonly used in online and physical stores. The Nilson survey projected the growth of purchase transactions worldwide for the six regions and reported “Among the global brand cards, those issued in the Asia-Pacific region generated 102.50 billion purchase transactions in 2017, accounting for 34.68% of the world wide purchase transaction total. This surpassed the United States for the first time. The Asia-Pacific is projected to reach 54.03% of the worldwide total by 2027” [1]. As the Card transaction increases the rate of fraudsters will also grow concurrently. Credit Card Fraud Detection (CCFD) is a challenging research undergone by the research community as the fraudsters change their behavioral pattern now and then which becomes an alarm for the banks to set a solution. The fraud damage is found after the fraudulent transaction is completed. Forecasting the fraud before it happens helps to prevent loss.

As the fraudsters increase, the technology for solving the problem plays a vital role. Any irregular pattern in the purchase transaction is considered as doubtful and further action is taken [2]. Machine Learning has various algorithms to overcome CCFD as it is

a binary classification problem, having the majority of normal transactions and the minority of fraudulent transactions. As the dataset is highly imbalanced when applying machine learning directly to the dataset, prediction inclines towards the majority class [3]. The important challenges faced by CCFD are the class imbalance - The nature of the data have less fraudulent behaviors than the legitimate [4], Concept drift - is due to the spending behavioral changes of the cardholders due to different circumstances that are classified into two; active and passive adaptation [5] finally misclassification of data - The legitimate transactions may be considered as fraudulent and inversely, therefore causes false alarm rate [6, 7] which create inconvenience to the cardholders.

The remainder of this paper is structured as follows. The following Sect. 2 addresses about the main challenge of class imbalance, Sect. 3 describes the CCFD techniques by different machine learning approach, Sect. 4 discusses the measures and metrics evaluated in CCFD of other research papers, Sect. 5 explains the Dataset and Tools used by other researchers, and Sect. 6 concludes with observations.

2 Class Imbalance

One of the challenging phases in CCFD is to train the classifier as the data have a minimum ratio of fraud, may lead to misclassification of data.

Preprocessing the data will balance and reduce the noise between the binary classes [8]. Therefore original dataset is balanced by performing oversampling to increase the fraudulent class or undersampling to remove the legitimate class [9], studies reveal that the performance of the classifier is better when oversampling technique SMOTE (synthetic minority oversampling technique) is used [3] to handle class imbalance. The two metrics; Balanced Classification Rate (BCR) and Matthews Correlation Coefficient (MCC) was applied and found FraudMiner performs well in handling class imbalance when compared with other classifiers. In [10] SMOTE increased the accuracy of the classifier by 2–4%. Though adaptive synthetic sampling (ADASYN), ranked minority sampling (RAMO) were recently suggested [11] but lead to a problem of class distribution when the number of iteration is high, hence the researchers suggest ensemble classifiers will have better results than a single classifier with unbalanced data.

Akila and Reddy [12] proposed an Overlapped-Majority Bagging method to handle class imbalance by selecting 60% data of minority class into a single bag with few majority data, without creating an internal imbalance in the bag. Awoyemi et al. [13] approached a hybrid sampling technique where the genuine class is undersampled and the fraud class is oversampled with the same number of iterations. Wang and Han [14] expressed an integrated clustering technique by taking random samples to improve the classification of imbalanced data where the SVM model made higher accuracy of minority and majority class [14]. Van Vlasselaer et al. [15] applied weighting method for neural networks and logistic regression and subsampling were used in random forests. The disadvantage of SMOTE is presented in [3] as it produces noise and chance of overlapping among classes and create overfitting and revealed undersampling is better than oversampling. Jurgovsky et al. [16] built an account based undersampling technique by randomly picking a legitimate account from the set of a compromised account by repeating the process to create training dataset. CLUE [4] employed

undersampling for legitimate sessions by random skipping. The undersampling method effectively handle class imbalance problem [17–20]. Carneiro et al. [21] inferred that balanced dataset using undersampling majority class did not show a remarkable performance over unbalanced dataset when applied into a classifier, most of the classifiers do not classify the minority class in the dataset as the classifiers are biased by expressing the majority class.

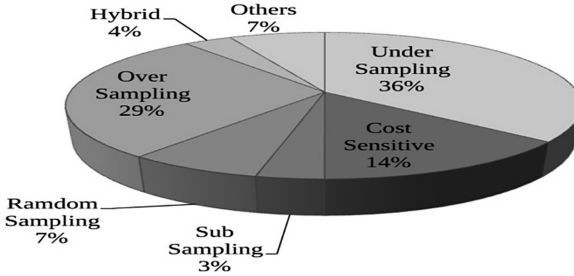


Fig. 1. Frequency of sampling techniques used in the literature.

Carneiro et al. [4] and Wang et al. [21] experimented random sampling to handle the data imbalance. Decision tree algorithm reduces the variance and overfitting of the dataset by adjusting the trees [22]. Clustering tree model [23] filters the majority class data based on the characteristic of the dataset considering class separable. Cost-sensitive methods solve the misclassification of data and made the classifier to consider the minority class [24–27].

3 Machine Learning Techniques of CCFD

The present techniques are based on supervised or unsupervised learning where attributes of the previous transaction are used in the training phase in supervised learning while unsupervised methods use clustering algorithms to group the customer behavior into fraud and legitimate patterns [28]. New algorithms are proposed by the researchers and some introduced hybrid techniques by combining the existing methods. An ensemble learning approach generated better performance when compared with other methods. The recent papers concentrate on Deep learning where the data should be huge in size.

3.1 Supervised and Unsupervised Learning

Supervised learning has labeled dataset to train the classifier. The samples of fraud and legal database are associated with their labels whereas unsupervised learning does not rely on labeled data to construct a model and does not need previous knowledge to train the model [6, 7]. Unsupervised learning extracts the significant features for data, [29] leveraging the accessibility of unlabeled data and adds dependency between data for training. Researchers proposed new methods and compared with the existing methods

and found the classifier producing a good result. Few papers suggested a hybrid approach of the existing classifiers with a comparison report. The following are the various techniques proposed by the researchers using supervised and unsupervised learning algorithms.

Seeja and Zareapoor [30] proposed Fraud Miner using frequent itemset mining to identify legal and fraud transaction pattern of the customer in training stage by applying Apriori algorithm which returns the frequent itemset with more number of occurrences of legal and fraud patterns then proposed a matching algorithm during the testing phase to traverse the pattern database and match the new incoming transaction to detect fraud or legal. Hegazy et al. [31] created Frequent Pattern based on customer's previous transaction activity as Legal or Fraud transactions introducing using Rough Set and Decision Tree Technique clustering algorithm in Enhanced Fraud Miner algorithm which attained good improvement in finding the false alarm rate when compared to other models. de Sá et al. [24] proposed Fraud-BNC to generate Hyper-Heuristic Evolutionary Algorithm (HHEA) for dataset classification and it discovers the legitimate transaction when the class probability is less than 0.5 when compared with other methods the chance of missing a fraud class can be reduced by increasing the probability of fraud transaction. Van Vlasselaer et al. [15] suggested APATE (Anomaly Prevention using Advanced Transaction Exploration) used the RFM framework (Recency - Frequency - Monetary Value) based on the geographic location of the customer and achieved high AUC score and accuracy particularly for random forest method which exactly chooses the first transaction cycle of fraudulent.

Dai et al. [32] combined supervised and unsupervised approach by fusing various models to train and record the spending behavior for each cardholder based on their previous transactions and for every new transaction the fraud score is computed from the fraud pattern. Batani [33] designed a hybrid and adaptive method to detect fraud using cardholder's financial status, social status, and OTP by assigning weights using Artificial Neural Networks to produce card holder's social status and Hidden Markov Model (HMM) to extract financial profile from bank database. Behera and Panigrahi [34] combined fuzzy clustering and neural network where Fuzzy c-means clustering model groups the related dataset and neural network decrease the misclassification rate based on the parameters purchase amount, type and time. Jain et al. [35] presented a hybrid method to detect fraud by Rough Set and Decision Tree Technique where the mathematical tool Rough Set used to preprocess the data, a decision tree is built and applied to each row in the database and classification is acquired. Kamaruddin and Ravi [36] proposed a combined model of particle swarm optimization (PSO) and auto-associative neural network (AANN) to find an output on one-class classification (OCC) in the big data model using SPARK cluster and attained 89% true classification.

Santos and Ocampo [37] suggested Naïve Bayes method with clustering technique and produced accuracy by 81% using 3 clusters and an average accuracy of 79% with 2 clusters, recommended increasing the number of clusters to give the finest initialization of EM clustering model reporting the behavioral pattern of cardholders and fraudsters. The accuracy is reliant with the huge transaction of the fraudsters with the implication of high True Positive Rate (TPR) and overall accuracy. Hassan [38] conducted research by assigning different costs for false negative (FN) using the cost matrix and applied various cost-sensitive classifiers and investigated the performance by applying Bayes

minimum risk. Yee et al. [39] tested the metrics for five Bayesian classifiers and reported that Bayesian classifiers are notably better after applying the Principle Component Analysis (PCA) method for the data to discard irrelevant attributes. Awoyemi et al. [13] compared Naïve Bayes, K-nearest neighbor, and Logistic regression models and concluded that K-NN performs well. Nur-E-Arefin and Islam [40] compared 16 classifiers to detect credit card fraud and reported that Multi-Class Classifier performs better.

Tran et al. [41] presented two real-time data-driven approaches, one class support vector machine (OCSVM) and T2 control chart which attained good accuracy and low false positives. Askari and Hussain [42] initiated Fuzzy-ID3 classifier to split the attributes possessing the highest information gain and the leaf nodes categorize the transactions as fraud or legitimate. Artikis et al. [43] introduced a prototype model SPEEDD for proactive event-driven decision-making for creating automated fraud pattern using the parameters country, merchant, amount and customer, applied OLED (Online Learning of Event Definitions) a statistical model to learn the pattern for a small set of data.

3.2 Ensemble Learning

Ensemble learning is supervised learning where many classifiers are combined and applied to make predictions. As there is an increase in computational power to train the classifier in an appropriate time frame can be a good method for CCFD [44].

Saia and Carta [9] proposed a novel method using Fourier transform and Wavelet transform to analyze and experiment the data by Random Forests method with k-fold cross-validation and compared with ten Data mining algorithms and concluded Random Forests as the best performing method. Dal Pozzolo et al. [8] compared Random Forests (RF) with Neural Network (NN) and Support Vector Machine (SVM) where the Random Forests performed well as expected and suggested the accuracy can be improved by increasing the training data size. Carneiro et al. [21] applied 10-fold cross-validation to Random Forests, SVM and Logistic regression and tested with balanced and unbalanced data then confirmed that Random Forests attained the best performance. Two techniques of Random-tree-based random forest and CART are applied [17] to train the dataset for legal and fraud pattern then compared their performance, voting mechanism inferred that both have equal weight. Noghani and Moattar [11] proposed an ensemble classification with cost-sensitive decision trees and increased the performance by 1.8% to 2.4%. Nami and Shajari [25] forwarded two stage novel approach using KNN and Dynamic Random Forest (DRF) by allotting weight to recent transactions can increase the performance of the system; Tree ensemble learns the pattern from both legal and fraud class based on their hierarchical structure which is more popular in solving problems having unbalanced data. Patil et al. [45] designed a framework for the fraud detection and prediction and concluded Random Forest Decision Tree model is best and the only difficulty is overfitting when data size expands. Wang et al. [3] proposed ensemble framework partitioning and hierarchical clustering (RFPH) with partitioning by Random Forest and clustering by C4.5. In [46] three classifiers are compared and pointed random forest performs well than the logistic regression and decision tree.

Zhang et al. [47] proposed stacking ensemble learning for each set of data to train different learner, the result produced by the first learner is considered as features to train meta-learner. Zareapoor et al. [22] compared bagging ensemble classifier with Naive Bayes, K-Nearest Neighbor, Support Vector Machines (SVM), and Ensemble classifier outperforms its competitors. Wang and Han [14] built an ensemble AdaBoost integrated learning algorithm having SVM as the base classifier and used cluster analysis for imbalanced dataset. Su et al. [48] implemented 10-fold cross-validation and suggested XGBoost as the best model, the output of several ensemble models revealed AUC over 0.9000 in MCC (Merchant Category Code) misuse. Kumari and Mishra [10] concluded that ensemble classifier performs well than any other classifier.

3.3 Deep Learning

Deep learning is one of the class in machine learning [49] which is based on artificial neural networks. In deep learning, the output of each level transformed to be an input data into a more abstract and composite representation.

Fu et al. [27] recommended Convolutional Neural Networks (CNN) framework to identify the fraud behavioral pattern and compared with SVM, NN, RF which exhibited CNN performs better than others. Jurgovsky et al. [16] proposed LSTM (Long Short-Term Memory) to aggregate previous purchase pattern of the cardholder and to improve the accuracy of the incoming transaction, compared sequence learner LSTM and static learner (Random Forest) where LSTM is prone to overfitting even with few nodes, hence suggested to increase the size of data. CLUE [4] was embedded by two deep learning methods Item2Vec and RNN including LSTM as default, the learning is based on the details of items browsed, and the time domain of browsing behavior. CLUE performs better than traditional methods. Roy et al. analyzed four deep learning method ANN, RNN, LSTM and Gated Recurrent Units (GRUs) using 10-fold cross-validation and the performance of GRU is considered as best with the accuracy of 0.916 and concluded ANN was the worst with accuracy score 0.889 [19]. Kim et al. [26] compared and confirmed Hybrid Ensemble with deep feed-forward Discriminative models to build multiple layers of abstraction, experimented with deep neural network which produced better performance. The 10-layer deep Variational Auto-Encoder (VAE) was applied in [51] and compared with Decision Tree, SVM and Ensemble Classifier (AdaBoost algorithm) where AdaBoost achieved high Precision and recall for VAE. Pumsirirat and Yan [29] presented Auto-Encoder and Restricted Boltzmann and confirmed supervised learning is appropriate for the historical transaction in credit card fraud detection. Rushin et al. [18] compared Deep Learning, Logistic Regression, and Gradient Boosted Tree, based on the AUC values of the feature set concluded deep learning is the best performer and LR (Logistic Regression) is the poor performer. Niimi [52] compared Deep learning with other supervised algorithms and confirmed the accuracy produced by Deep learning is close to Gaussian kernel SVM.

Table 1. Usage frequency of machine learning techniques in CCFD.

Machine learning	Methods	Usage frequency	References	Advantage	Disadvantage
Supervised Learning	Neural Network	10	[8, 9, 15, 17, 20, 27, 32–34, 38]	Highly accurate and reliable	Need to understand and label the input More computation time required for the training phase
	Support Vector Machine	16	[4, 8, 14, 17, 20–24, 27, 30, 38, 41, 48, 52, 53]		
	Bayesian Network Classifiers	15	[4, 9–11, 13, 17, 22, 24, 30, 32, 33, 37–40]		
	K-Nearest Neighbor	5	[9, 13, 22, 25, 30]		
	Logistic Regression	14	[4, 9, 13, 15, 20, 21, 23, 24, 26, 32, 38, 45, 46, 53]		
	Decision Tree	12	[3, 9, 11, 17, 20, 23, 26, 32, 38, 45, 46, 48]		
Unsupervised Learning	Expectation-Maximization	1	[37]	Easy to find unknown patterns and features of data	Computationally complex Less accurate due to unlabeled input
	K-Means	3	[22, 37]		
	Fuzzy C-Means	1	[34]		
	DBSCAN	1	[32]		
	Hidden Markov Model (HMM)	3	[2, 32, 33]		
	Self-Organizing Map (SOM)	1	[32]		
Ensemble Learning	LINGO	1	[31]	Avoid the overfitting problem and gives better predictions when compared with a single model	Computation time is high Reduces model interpretability due to increased complexity
	Random Forest	20	[3, 4, 8–10, 15–17, 20, 21, 23]–[25, 27, 30, 38, 45, 46, 48, 53]		
	Boosting	7	[9, 18, 20, 23, 48, 52, 53]		
	Bagging	3	[10, 12, 22]		
Deep Learning	Voting	1	[10]	No need for feature extraction and labeling of data	A large amount of data is needed to find the pattern Create overfitting problem in the model
	Stochastic Gradient Descent	1	[9]		
	Long short-term memory	3	[4, 16, 19]		
	Deep Feed Forward NN	3	[18, 26, 51]		
	Variational Autoencoder (VAE)	1	[51]		
	Auto Encoder (AE)	2	[29, 41]		
	Restricted Boltzmann Machines	1	[29]		
	Recurrent Neural Network	2	[4, 19]		
	Convolutional Neural Network	1	[27]		
	Generative Adversarial Network	1	[50]		

4 Classification Metrics for CCFD

Evaluation metrics identify the performance of the classifier when applied to the dataset. A 2×2 Confusion matrix [53] gives clarity by comparing the predicted and actual class. Based on the survey we found 20 evaluation metrics out of 41 papers. Then we summarized 12 metrics which are most commonly used by the researchers.

True Positive Rate (TPR). TPR is otherwise called as Sensitivity or Recall [54] is the number of actual positives which are predicted positive, in CCFD the actual positives are fraudulent transactions.

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (1)$$

True Negative Rate (TNR). Also called specificity, it is the proportion of actual negatives which are predicted negative, recognize how well the classifier identifies the legitimate transactions.

$$\text{TNR} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (2)$$

False Positive Rate (FPR). The proportion of legitimate samples that were wrongly predicted as fraud [55, 56].

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \quad (3)$$

Accuracy (Acc). The proximity measure of exactly classified samples [57] to the total number of instances.

$$\text{Acc} = \frac{\text{TN} + \text{TP}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}} \quad (4)$$

Precision. Precision or Positive Predictive Value represents the proportion of positive samples that were correctly classified to the total number of positive predicted samples [56].

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (5)$$

F-measure. Otherwise referred as F1-score, and it represents the harmonic mean of precision and recall. The value ranges from 0 to 1, if the value is high then *F*-measure indicates high classification performance [56].

$$\text{F-measure} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

Area Under the Curve (AUC). Represents the capability of distinguishing between classes. When AUC is higher the capability of prediction is better, the value bounds between 0 and 1.

Receiver Operating Characteristics (ROC). The receiver operating characteristics curve is a two-dimensional graph in which the TPR represents the y-axis and FPR is the x-axis at various threshold settings [56].

Balanced Classification Rate (BCR). The average of sensitivity and specificity [30]

$$\text{BCR} = \frac{1}{2}(\text{TPR} + \text{TNR}) \quad (7)$$

Matthews Correlation Coefficient (MCC). Represents the correlation between the observed and predicted classifications. MCC value is between $[-1, 1]$, if the coefficient is +1 indicates a perfect prediction, -1 represents prediction is exactly wrong [56].

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}} \quad (8)$$

Geometric Mean (GM). The Geometric Mean (GM) aggregates sensitivity and specificity [56] measured at one specific threshold.

$$\text{GM} = \sqrt{\text{TPR} \times \text{TNR}} \quad (9)$$

Cohen's Kappa Coefficient (K). Measures the chance of an agreement between the predicted and the actual classes [53], given by

$$K = \frac{P_0 - P_e}{1 - P_e} \quad (10)$$

where P_0 is the observed agreement, and P_e is the expected agreement.

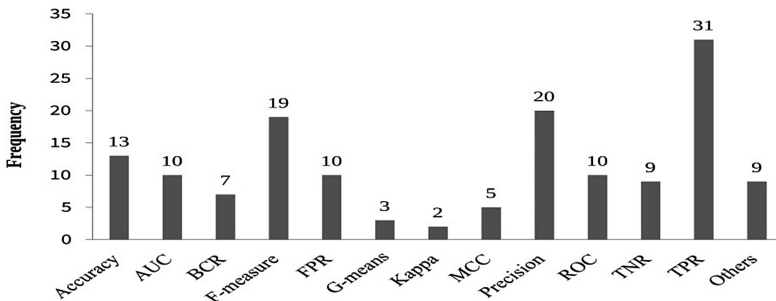


Fig. 2. Frequency of evaluation metrics used in the literature.

Precision is used to find the good classifier, [20] When the precision value is high it indicates the classifier is good. Accuracy [22, 30] being a common metric, it is not appropriate for CCFD due to data imbalance may lead to misclassification. The performance of the classifiers shows good results with preprocessed dataset than the raw dataset [46]. As Fig. 2 indicates that most of the papers have experimented Sensitivity (31), Precision (20), and F-measure (19) performance metrics to measure the efficiency of CCFD.

5 Dataset and Tools for CCFD

The publicly available dataset can be classified as Credit approval, Default payment, and Transactional dataset. The attribute class or default is the indicator of prediction with binary value 0(legitimate) and 1(fraud). Even in a very huge transactional dataset, the occurrence of fraud is very minimal which creates an imbalance problem. Most of the real data have a ratio of 99% of legal transaction and 1% of them are fraud [3]. Table 2 visualizes the experimental datasets used by the researchers in the field of CCFD. Few of the confidential real data is from the bank of China, Latin America, Singapore, South Korea, etc.

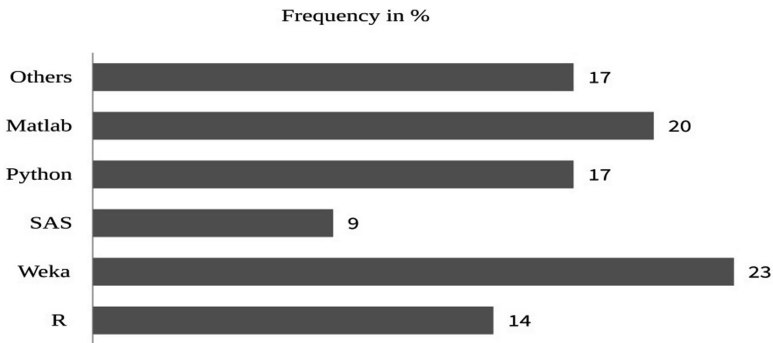


Fig. 3. Occurrence of software tools in the literature.

Figure 3 depicts Weka, Matlab, python are popularly used software tools to develop the CCFD system. 31% of the publication have not mentioned about the software used. When the research is based on big data analytics, there will be a huge volume of data which can be implemented in Apache Hadoop, Spark, etc. Tensorflow, H2O, Pytorch, Keras, etc. are the libraries imported in the application of deep learning.

Table 2. Dataset used for CCFD.

Dataset	Number of instances	Fraud instances	References	Source
UCSD-FICO DataMining Contest 2009	100,000	2293	[12, 22, 30, 38, 40]	https://www.cs.purdue.edu/commugrate/data/credit_card/
European – Credit card transaction	284, 807	492	[3, 9, 13, 23, 29, 41, 46, 50, 51]	https://www.kaggle.com/mlg-ulb/creditcardfraud
German–Credit	1000	300	[9, 10, 29, 35, 45]	UCI Machine Learning Repository
Taiwan–Default Credit Card	30000	6636	[20]	https://www.kaggle.com/uciml/default-of-credit-card-clients-dataset
Australian-Credit Approval	690	307	[10, 29, 52]	https://data.world/uci/credit-approval
cc Fraud	1048575	62739	[36]	https://packages.revolutionanalytics.com/datasets/
Confidential data from bank			[4, 8, 11, 12, 15, 17–19, 21, 23, 24–27, 42, 47, 48]	
Synthetic data			[33, 34, 43]	
Simulated data			[31, 32, 37, 39]	

6 Conclusion

The review concludes that Supervised Learning is widely used for CCFD than the unsupervised. We observed that the most commonly used fraud detection techniques are Support vector machine, Bayesian classifiers, Logistic Regression, and Decision Tree. It was also suggested that the performance of Hybrid methods are better than using a single classifier. Table 1 express that Ensemble learning showed good performance with Random Forest and the usage frequency is more when compared with other techniques. Although the recent researches are based on deep learning in predictive analytics, the performance when compared to supervised learning is still in the research process. Based on the study it was observed that Deep learning can perform well on image data than numeric. We observed that 55% of publications have used sampling method to handle the class imbalance and few have compared the performance using raw data and sampled data, where 45% have not reported about any sampling technique. From Fig. 1 it was noted that many of the research articles have used the undersampling technique. It was also spotted from Fig. 2 that Sensitivity, Precision, and F-measure have produced good results for the performance analysis in

CCFD and also suggested that the performance can be improved by increasing the size of the data.

In future cloud analytics becomes a popular analytical tool to extract information from massive data, implementing Machine learning with cloud or fog computing environment paves a way for better performance.

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